

Whenever There Is A **MECHANICAL MOVEMENT** That Has To Be Counted Or Detected, A **REED SWITCH** Is **APPLICABLE**

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Standard reed switches are small, have high reliability and a long life for demanding applications in automotive, industrial and all kinds of position control, position-sensing applications. Standard versions can be modified with terminals according to specific needs. Kai Hold, marketing manager, PIC GmbH, in a conversation with Deepshikha Shukla, explains how reed switches help in motion-sensing devices and their parts

Q. Why are reed switches needed? What are their types?

A. Reed switch is a component for non-touch switching for handling loads from almost 0W to 250W in a wide range of applications. There are four types of reed switches: micro, standard, power and special. Micro switches are the smallest and are available in various sensitivity ranges. These find applications in hearing aids and implantable medical devices.

Q. How does an interactive tool help select actuation type of a reed switch?

A. The interactive tool enables one to simulate different methods of actuating a reed switch. It helps learn about its switching characteristics. This enables selection of actuation type among one pole, parallel, rotation, and shielding, and mode among auto and manual.

Q. Can reed switches help go green?

A. A reed switch consists of glass and metal and, hence, is easy to recycle. It consumes no power in standby mode and does not require any separate power supply.

The hermetically-sealed reed switch does not produce any micro-emissions like ozone or nitrogen-oxide. It is immune to electrostatic discharge (ESD), so no additional safety components and shielding arrangements are required either.

While a semiconductor often becomes perceptibly hot, a reed switch remains cool and reduces measures required for cooling like using extra heat-sink(s).

Q. How does a reed chain work?

A. Reed chains allow simple and

effective continuous measuring of liquid levels; for example, big water-purification systems, heating systems and so on. Reed switches and resistors mounted onto a PCB are collectively called reed chains.

A reed chain is inserted into a long, matching float that moves up and down the stem, according to level of the liquid. Total resistance changes with the magnet inside the float, switching the reed switch on and off. This change in resistance provides easy, seamless and continuous level information.

Level sensors used for water/oil tanks and industrial ink-printing machines, among others, combine the switching element and magnet into one unit.

Q. What are the dos and don'ts of handling reed switches?

A. Reed switches offer high shock resistance up to hundred grams, but a drop on a hard surface can generate more shock, which can lead to de-adjustment of contacts. Switches that have been dropped should be re-tested for sensitivity before usage.

Incorrect bending or cutting of terminals may lead to cracks in the sealing area due to heavy mechanical stress. To avoid this problem, remaining part of the terminals between the glass body and point of bending should be gripped tightly.

Q. Can reed switches be customised?

A. Dimensioning should be made with various sensitivity ranges like standard switches, power switches with contact ratings up to 250W and 5A, and latching switches with breakdown voltage up to 15,000V DC

for high-frequency circuits. Customer-specific, tailor-made sensing solutions take assembly, electrical parameters and design needs, like switching distance and environment of application, into consideration.

Q. What are the applications of reed switches?

A. Whenever there is a mechanical movement that has to be controlled or checked, like a magnet that moves with the water level, it activates a reed switch or reed sensor.

Typical industries using such products are home appliances, automated machines like coffee makers, tea makers, water-purification systems, washing machines, dishwashers, refrigerators and ACs.

In the industrial automation sector, these are used for position sensing for pneumatic or hydraulic cylinders, and in a variety of safety functions like CNC or laser-cutting machines, where when the door is opened, the machine stops immediately.

The Internet of Things (IoT) and home automation are great opportunities for manufacturers of sensors because we need them everywhere. Water and gas metering are also big markets. We assume that electric vehicles (EVs) like electric bikes, electric bicycles, speed sensors for fitness equipment like treadmills powered by battery, etc will increase, and all of them will need speed measuring. All battery power applications are perfect applications for reed-switch-based sensors due to low power consumption. Hence, whenever there is a mechanical movement that has to be counted or detected, a reed switch is applicable. **EFY**